

DESIGN AND DEVELOPMENT OF AI-ASSISTED DRONE FOR HUMAN DISTRESS DETECTION

B.Tech.

Project Work 1

Department of Electronics and Communication Engineering



Project Team

Sl. No.	Reg. No.	Student Name
1	21ETEC004016	BRIAN CHARLES
2	22ETEC004011	BHANU SIMHA REDDY
3	22ETEC004048	KIRAN SHINDE
4	22ETEC004051	SIMHADRI THANMAI

**Supervisor : Mr. Viswanath K,
Assistant Professor, Department of ECE**

January – 2026

**FACULTY OF ENGINEERING AND TECHNOLOGY
M. S. RAMAIAH UNIVERSITY OF APPLIED SCIENCES
Bengaluru-560054**

FACULTY OF ENGINEERING AND TECHNOLOGY



Certificate

*This is to certify that the Project Work titled “**DESIGN AND DEVELOPMENT OF AI-ASSISTED DRONE FOR HUMAN DISTRESS DETECTION**” is a Bonafide record of the work carried out by **Mr. BRIAN CHARLES** , Reg. No. **21ETEC004016** , **Mr. BHANU SIMHA REDDY**, Reg. No. **22ETEC004011**, **Mr. KIRAN SHINDE** , Reg. No. **22ETEC004048**, **Ms. SIMHADRI THANMAI**, Reg. No. **22ETEC004051**, in partial fulfilment of requirements for the award of **B.Tech** Degree of M. S. Ramaiah University of Applied Sciences in the Department of **Electronics and Communication Engineering***

January – 2026

Supervisor: Mr. Viswanath K
Assistant Professor, Department of ECE

Dr. T. Christy Bobby

**Professor and Head,
Department of ECE**

Dr. Sarat Kumar Maharana

**Dean,
FET, MSRUAS**

Declaration

DESIGN AND DEVELOPMENT OF AI-ASSISTED DRONE FOR HUMAN DISTRESS DETECTION

The project work is submitted in partial fulfilment of academic requirements for the **B.Tech** Degree of M. S. Ramaiah University of Applied Sciences in the Department of **Electronics and Communication Engineering**. This project work is a result of our own work and is in conformance to the guidelines on plagiarism as laid out in the University Student Handbook. All sections of the text and results, which have been obtained from other sources are fully referenced. We understand that cheating and plagiarism constitute a breach of University regulations, hence this project report has been passed through plagiarism check and the report has been submitted to the supervisor.

Sl. No.	Reg. No.	Student Name	Signature
1	21ETEC004016	BRIAN CHARLES	
2	22ETEC004011	BHANU SIMHA REDDY	
3	22ETEC004048	KIRAN SHINDE	
4	22ETEC004051	SIMHADRI THANMAI	

Date: 30 January 2026

Acknowledgements

We take this opportunity to express our sincere gratitude and appreciation to Ramaiah University of Applied Sciences, Bengaluru for providing us an opportunity to carry out our project work.

It brings a sense of privilege to associate this work with Mr. Viswanath K, our project supervisor. Working under his guidance has been a precious opportunity. We are immensely grateful for his vigilant supervision, constant encouragement, inspiring ideas, constructive criticism, and timely expertise, which reflects in the final outcome of this work.

We are grateful to the University and thank the Head of Department of Electronics and Communication Engineering, Dr. T. Christy Bobby for providing us with this opportunity, guidance, and encouragement throughout.

We would like to earnestly thank Dr. Sarat Kumar Maharana, the Dean, Faculty of Engineering and Technology, for facilitating academic excellence in the college that helped in completing this project. We are also thankful to Prof. Kuldeep Kumar Raina, the Vice Chancellor, for facilitating academic excellence in the college that helped in completing this project.

Furthermore, we take this opportunity to acknowledge and appreciate the support we got from our mentors, faculty members, Lab Assistants for providing all the facilities and resources for the successful completion of the project

Finally, we are thankful for the constant support shown by our family, friends and well-wishers, their unwavering support has been a source of motivation throughout this journey

Abstract

Today, unmanned aerial vehicles, more commonly known as drones, can be equipped with high-resolution cameras and systems powerful enough to provide effective and efficient aid to Search-and-Rescue (SAR) operations in any natural or man-made catastrophe environments these result in huge loss of human life and property. In such cases, rescue operations to find distressed individuals will be challenging due to the geographical areas and hazardous environment and delayed location reporting

This project is a prototype that can help the SAR/rescue teams in locating victims, who may be in distress, as quickly as possible for a chance of survival. Therefore, using drones with computer vision can increase the detection rate while reducing rescue time, reduce risks, can travel through dangerous paths and send the data in Real-time

This system integrates a Raspberry Pi 5 with a Pixhawk 2.4.8 flight for manual flight and situational awareness. Unlike existing drone-based detection which relies on manual observing the data, this system utilizes YOLOv8-Pose estimation to analyse 17 human key points to distinguish between "Safe" and "Distress" states by calculating the vertical displacement of wrists relative to shoulders with a 90% success rate. Upon confirming a distress gesture, the system extracts live GPS telemetry via the MAVLink protocol, captures a visual snapshot of the victim, and dispatches an alert including a accessible Google Maps link to a remote ground station via the Telegram Bot API. A real-time VNC-based video feed provides the manual pilot with a tactical "Heads-Up Display" (HUD) for continuous surveillance.

Table of Contents

Certificate.....	(i)
Declaration.....	(i)
Acknowledgements.....	(ii)
Abstract	(iii)
Table of Contents.....	(iv)
List of Tables.....	(vii)
List of Figures.....	(viii)
Nomenclature.....	(ix)
Abbreviations and Acronyms.....	(x)
 Chapter-1: Introduction.....	 01
1.1 Significance of Drones in Image-Based Monitoring.....	02
1.2 Motivation and Need for the Project.....	02
 Chapter-2: Background Theory	 03
2.1. Existing Research and Technical Development	03
2.2 key Principles and Assumptions.....	03
2.2.1 Major Assumptions.....	04
2.3 Key Definitions.....	04
2.4 Merits, Demerits and Applicability.....	05
2.5 Techniques for Human Detection.....	05
2.5.1 Gesture/Pose Detection Methods	06
2.5.2 Challenges in Real-Time Detection Systems	06
2.6 Research Gaps	07

Chapter-3: Aim and Objectives.....	08
3.1Title.....	08
3.2Aim.....	08
3.3 Objectives	08
3.4 Methods and Methodology.....	09
 Chapter-4: Critical Thinking	
4.1 Hardware Selection and Component selection	10
4.2 Engineering Calculations and Propulsion Analysis.....	11
4.2.1 Weight Budget Estimation (All-Up Weight).....	11
4.2.2 Thrust-to-Weight Ratio Calculation.....	11
4.2.3 Battery C-Rating and Current Safety Check	12
4.2.4 Battery Safety.....	12
4.2.5 Theoretical Flight Time Calculation	12
4.3 Hardware Integration and Wiring Strategy	13
4.3.1 Motor Mapping	13
4.4 AI Vision Pipeline Implementation.....	13
4.4.1 The Detection Logic	13
4.5 Advantages of Raspberry Pi.....	14
4.6 Communication System (Telegram Bot).....	14
4.7 Validation Scenarios.....	14
 Chapter-5: Results and Discussions	
5.1 MAVlink Connection	15
5.2 Drone Platform/Prototype.....	16
5.3 Ground Station	17
5.4 Drone Arming and Flight.....	17
5.5 Human Detection.....	18
5.6 Drone Flying/Testing	19
5.7 Real-Time Human detection from the drone.....	20
5.8 Telegram Output.....	22

Chapter-6: Project Costing	24
Chapter-7: Conclusion and Future Directions	25
7.1 Conclusion	25
7.2 Comparison between existing model vs our prototype	25
7.3 Accuracy	26
7.4 Future Directions	26
References	27
Appendix	28

List of Tables

Table 3.4 Methods and Methodology.....	09
Table 4.1 Hardware Architecture and Component Selection.....	10
Table 4.2.1 Weight Budget Estimation (All-Up Weight).....	11
Table 5.2 Drone Platform/Prototype.....	16
Table 5.8 Telegram Alert/Output.....	23
Table 6.1 Project Costing.....	24
Table 7.2 Comparison between existing model vs our prototype.....	25

List of Figures

Figure 1.1 (A) A Person in Distress.....	01
Figure 1.1(B) Human Distress Detection Drone.....	01
Figure 2.5 comparison between YOLO family.....	06
Figure 5.1 MAVlink Connection.....	15
Figure 5.2.1 Drone Top View.....	16
Figure 5.2.2 Drone Side View.....	16
Figure 5.2.3 Drone Bottom View.....	16
Figure 5.3 Ground Station.....	17
Figure 5.4 (A) Drone Arming.....	17
Figure 5.4 (B) Arming check low RPM Take off.....	18
Figure 5.5.1 RPI&CAM	18
Figure 5.5.2 Human Detection	18
Figure 5.6 Drone Flying/testing.....	19
Figure 5.7 (A) people are present to test and validate the system.....	20
Figure 5.7 (B) Number of Humans Detected and are Safe.....	21
Figure 5.7(C) Humans Detected and in Danger	21
Figure 5.8.1 Telegram Alert.....	22
Figure 5.8.2 Google Link.....	23
Figure 5.8.3 Google Maps.....	23
Figure 7.4 Prototype Performance vs Altitude.....	26

Nomenclature

A	Acceleration (m/s^2)
F	Force (N)
T	Temperature (K) ,(C)
N	Speed (RPM)
DoF	Degrees of Freedom
W	Track Width (m)

Abbreviation and Acronyms

Abbreviation	Full-Form
AI	Artificial Intelligence
BLDC	Brushless Direct Current
CV	Computer Vision
COCO	Common Objects in Context
BOT API	Robot Application Based Interface
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
MAVlink	Micro Air Vehicle Link
UART	Universal Asynchronous RX/TX
UDP	User Datagram Protocol
UAV	Unmanned Aerial vehicle
Wi-Fi	Wireless Fidelity
NDMA	National Disaster Management Authority
RPI	Raspberry Pi 5
SAR	Search and Rescue

1. Introduction

Every year natural and man-made catastrophes like floods, cyclones, earthquakes, disasters, and fires and industrial mishaps take place. The consequences of these disasters result in huge loss of human life and property. In such cases, rescue operations to find distressed individuals will be challenging due to the geographical areas and hazardous environment. one of the greatest issues faced by rescue teams is safety, keeping both rescuers and victim's life in risks.

Drones have found their use in disaster scenarios since they are able to travel to hazardous and inaccessible locations and give real-time images and videos, it assists authorities to know more about the situation. Nevertheless, drone systems used by majority of people need constant human supervision and this is time consuming and can cause errors. To counter this drawback, smart systems are necessary that will be able to process visual information by automatic means and implement the Challenges.

The development and design of an AI Assisted Drone For Disaster Human distress detection is a prototype that will be deployed to help save lives more efficiently in such situations, by detecting distressed human beings and sending their status and GPS coordinates to the ground station (rescue teams) as shown in **Figure 1.1 (A) and 1.1(B)**



Figure 1.1 (A) A Person in Distress



Figure 1.1(B) Human Distress Detection Drone

1.1 Significance of Drones in Image-Based Monitoring

Image-based monitoring method is quite essential in identifying the existence of Humans in a natural and man-made catastrophes that maybe beyond human reach involving high risks.

In the event of a disaster, the use of drone is of great significance for disaster monitoring. They offer aerial images which cannot be viewed through the ground-based techniques. A drone has the capability to serve a large area in a limited time period and give real time images of the point of disaster, This assists the authorities to have visuals of the damages and organize rescue missions more efficiently.

This brings out the necessity of monitoring through images. Together with image recognition methods, it is also possible to develop systems capable of capturing images and, at the same time, analyse them intelligently. This project aims at enhancing the practicality of monitoring using drones through image recognition

1.2 Motivation and Need for the Project

long rescue operations usually arranged by local parties or NDMA, usually put their lives in danger when undertaking rescue operations. Human operators need to pay full attention to the video feeds of the drones and this makes them fatigued and they can overlook critical information.

With AI, drones can identify individuals on the spot, and emphasize on valuable information on sight. This minimizes manpower required by the human operators and increases the speed of response.

In this project, the drone does a surveillance of an affected area, where humans can be detected by image recognition, and basic gestures that can report whether the person is in distress or is safe. The system shows live video feed and gives useful information to the rescue teams(ground station), then it captures the frame of detected people from the same live video and is sent to the operator(ground station) via telegram containing time, date, number of people detected, GPS coordinates and a Google map link for a better directory

2. Background Theory

Some of the early approaches to human monitoring with the use of UAVs relied on more traditional computer vision protocols. At the same time, there were drones used in photography and general surveillance. experiments commenced on more specific uses in the context of human detection and interaction. it is explicitly indicated in these studies that the drones with cameras can observe and analyse human presence in aerial perspectives.

Nevertheless, the majority of the early works were aimed only at human detection as object and did not regard the analysis of the body posture or meaningful gesture acknowledgements.

2.1 Existing Research and Technical Development

In recent years, Advancements in drone Technology have significantly impacted the human World, particularly in videography and photography domain. A camera on a drone could help contributed much more than that, offering an efficient and cost-effective alternative for various large scale data collection to reduce human interaction

One of the earlier studies to explore the use of drone in human detection was conducted by Caputo, S., Castellano, G., Greco, F., Mencar, C., Petti, N. and Vessio, G Human detection in drone images using YOLO for search-and-rescue operations. Which assessed about unmanned aerial vehicles(UAV) know also as drones, can be equipped with high-resolution cameras and embedded GPUs powerful enough to provide effective and efficient aid to Search-and-Rescue (SAR) operations in remote and hostile environments. Locating victims, who may be unconscious or injured, as quickly as possible to improve their chance of survival, the findings from this research was to understand that the drones are capable of performing such tasks.^[1]

A research from Dong, C. and Du, G., 2024. An enhanced real-time human pose estimation method based on modified YOLOv8 framework. *Scientific Reports* that Surveillance, crowd management, and person tracking are major problems. And his research work describes the process of counting and identifying people in real-world, the take away from this research is about the raspberry pi and YOLOv8 model integration that achieves real-time person detection and pose estimation.^[2]

From the studies of Aliane, N., 2024. A Survey of Open-Source UAV Autopilots. we could understand various Open-source autopilots and protocols that are very useful for drones, the take away point from this research was PixHawk 4 and Ardupilot that primarily uses MAVLink, a lightweight messaging protocol that supports telemetry and control functions.^[3]

2.2 Key Principles and Assumptions

The aerial human gesture detection systems rely on a number of geometric as well as operational assumptions. In other words, the size of the human beings in the image is defined by the altitude of the drone and the resolution of its camera. Practically, the bigger the resolution of the camera, the higher the flight altitude, provided that we indulge too much in high flight or have a low-resolution camera, the pixels are small which is detrimental to detecting human beings.

2.2.1 Major assumptions

Drones' stability: it should be as stable as possible so that the camera does not get blurry products, its stability depends on several factors such as Central Gravity(CG) and aerodynamic disturbance.

- loiter mode: the drone should be stationary when an operation is in process, like the period of time the data-analytics may be perfect and it may be utilized in mapping an rescue strategy.
- environmental conditions: the drone should not be affected by environmental factors like the wind, turbulence etc.
- low light improvement method: the low light situation renders the image of the human being very hazy and unattainable thus improving the +video is an added advantage in the given project.
- High resolution: makes the detection from a far visual point easy, as it can clearly detect and analyse the key points in the captured frames.

2.3 Key Definitions

- UAV (drone): Is an unmanned flying vehicle that has cameras/sensors.
- Object Detection: The object recognition and classification problem in an image (persons in this case).
- Human Pose Estimation: Detecting important joints locations (elbows, Knees, etc) of a person.in the form of a 2D skeleton or heatmap
- Gesture Recognition: Categorizing a human body pose or movement into a redefined set Gestures here are restricted to a set of rescue gestures such as waving arms to draw attention.
- Real-Time System: This is a system that can autonomously handle video frame (or a video stream) at a sufficiently fast frame rate (typically 30 fps) to be interactive.
- System Throughput: frames per second (FPS) value that is obtained points to viability in real time. On board UAV systems are usually aimed at 10- 30 FPS based on equipment.
- Drone stability: Stability in this context, the fact that despite the effects on the environment and regardless of the weakness in the structural parts the drone remains at a stable position.
- loiter mode: is an automatic mode and it tries to keep the current location, heading and altitude.
- environmental conditions: There are a few major conditions that influence drone surveying, which include temperature, wind, humidity, rain, and clouds.

- image enhancement on low light: low-light image process is capable of raising image brightness and more accurately restoring colour and detail information.
- High quality: High quality drones refer to those drones that have the capacity to take detailed imagery, which are required in professional aerial photography, inspection, and surveying.

2.4 Merits, Demerits and Applicability

2.4.1 Merits: Aerial vision gives various situational awarenesses. Drones are able to reach isolated or unsafe areas and survey a greater area in a short period and provide an aerial perspective unattainable to ground station and rescue teams. They are able to travel tens of square kilometres per hour, and thus can search victims fast.

2.4.2 Demerits: the resources of UAVs are scarce. It can also be limited by battery life (usually 15-30 min flight time) to limit the time of operation. The changes in lighting, wind and rain and dust can have a negative impact on the input of the camera, making the video blur and jitter which can also be caused by UAV in motion. the visuals from the drone during flight detects the person as a tiny creature in the picture most of the time.

2.4.3 Applicability: The approach is very relevant to search-and-rescue despite being limited as follows. The research always underlines that drones are superior in aerial photography of the disaster area and victim search. Specifically, the use of signals through gestures may save lives of people in cases, when they could not move (were immobilized) and could not shout at rescuers. Human detection, which can be over the great distance and recognizing local gestures enable a drone to locate individuals, as well as comprehend their requirements e.g. grab their attention by waving.

2.5 Techniques for Human Detection

The detection of humans via Unmanned Aerial Vehicles (UAVs) has transitioned from simple manual video monitoring to highly sophisticated, autonomous systems. Modern techniques are primarily driven by Computer Vision (CV) and Deep Learning (DL), which allow drones to process visual data in real-time on lightweight, onboard hardware.

Aerial human detection is simply object detection applied in a drone image in relation to people. The prevailing paradigm is CNN-based detectors which have been trained on both labelled drone images. Leading method include:

- YOLO Family: The YOLO (You Only Look Once) by Redmon et al.^[4] and its variants (YOLO9000, YOLOv2, YOLOv3, YOLOv4, and so on) are in a widespread use. YOLO . It is capable of operating at extremely fast speed (real-time) and more than one person per picture. The take away from this source is YOLOv8, A computer vision model architecture for detection, classification, segmentation, and more.^[4]

comparing from the previous work of S. L. Krishna, G. S. R. Chaitanya, A. S. H. Reddy, A. M. Naidu, S. S. Poorna and K. Anuraj, "Autonomous Human Detection System Mounted on a Drone" ^[5], where they used a YOLOv4/standard CNN model, where the accuracy was moderate, hence we observed the comparison and went ahead with the YOLOv8 **Figure 2.1**

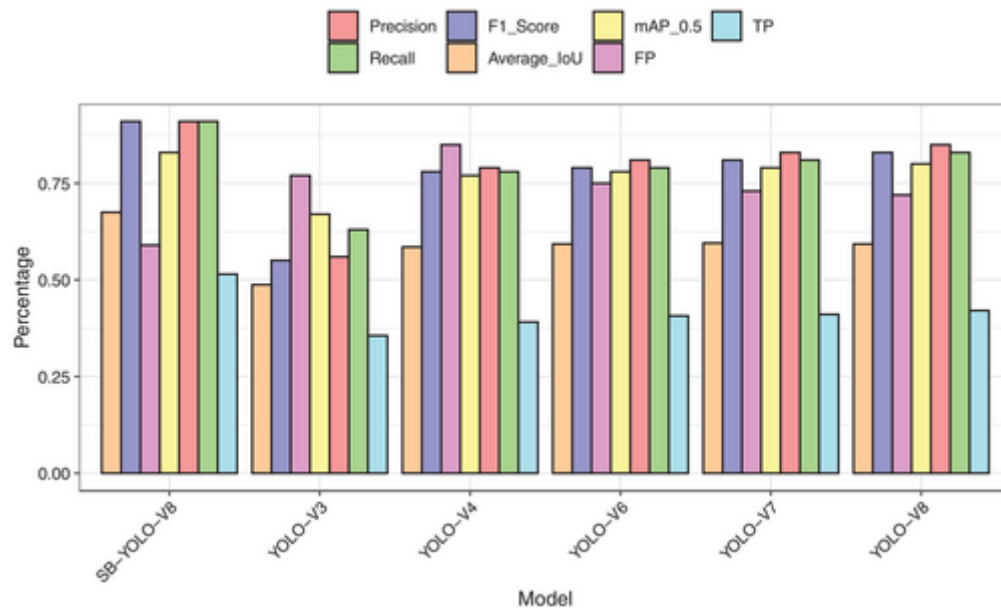


Figure 2.5 comparison between YOLO family

2.5.1 Gesture / Pose Detection Methods

- The other thing that needs to be done when the human is discovered is identifying their gesture or pose. Key methods include:
- Skeleton-Based Recognition: This method involves two steps, the initial step involves a pose estimator that obtains joint coordinates after which a pose is classified.^[2]
- Gesture recognition: There are studies that deal with gestures of the hands and not the entire body. Individually, as examples, hand-posture or sign-language recognition on a drone image involve viewpoint invariance, system should recognize the same gesture regardless of whether the individual is facing north or south and scale variations.^[2]

2.5.2 Challenges in Real-Time Detection Systems

UAV vision systems in real time have come across numerous issues:

- Small, Far Objects: Aerial imagery frequently its subjects are humans only a couple of dozens of pixels in size, and accuracy is difficult.

- This includes Fluctuating Weather: The outdoor images may be heavy in terms of shades, glare, or lack of contrast (dusk or nighttime). Quality also is degraded due to motion blur because of drone movement.
- Occlusion and Crowds: The UAV detectors that are currently in use are primarily trained on single subjects. When someone is in the way, or when a crowd is present, it follows that detection and pose-estimation is likely to fail. Last ground scenarios use multi-person pose estimation, whereas the aerial-view joints estimation is less advanced.^[3]
- Compute and Latency Constraints: It is difficult to show the real-time inference (30 fps or more) on a small drone.
- Limitations in the Data: the number of large aerial gesture labelled data sets is few.
- Numerous studies are therefore based on the gathering of data in a lab or model disaster location.
- False positives and Safety:.. the interpretation of a shadow as somebody waving. The systems should be very precise otherwise; they will give false data to rescue teams by producing unnecessary data . On the other hand, a missed person (false negative) may prove to be fatal.^[4]

2.6 Research Gaps

- The literature reviewed has given a number of gaps that stimulate future studies in the area of UAV based disaster rescue:
- Absence of Outdoor Gesture Data: The current gesture data of UAVs is almost indoors or controlled. No standard set of emergency rescue signs in the actual disaster situation is available. This lack of information inhibits training and benchmarking.
- Fixed Gesture Vocabularies: The existing systems presuppose a limited and stabilized system of gestures. Actual victims may adopt different or culturally- related cues.
- The existing works do not touch to the full extent on the recognition of unknown or new gestures.
- Small Object Detection: As has been mentioned, individuals are extremely tiny in high altitude UAV images.
- Real-Time Embedded Constraints: There are limited studies that are concerned with understanding the system-level constraints of real UAV hardware. Although on-board GPUs are mentioned frequently in the papers, the majority of experiments are performed in laboratory or offline.

3. Aim and Objectives

3.1 Title

DESIGN AND DEVELOPMENT OF AI-ASSISTED DRONE FOR HUMAN DISTRESS DETECTION

3.2 Aim

To design and develop an AI-assisted drone for real-time Human detection and their pose estimation.

3.3 Objectives

- 1 To carry out the literature review for human detection , pose estimation and Image processing Techniques and algorithms
- 2 To design and develop a drone platform integrated with Pixhawk flight controller and flysky with the Raspberry Pi 5 (16GB).
- 3 To Implement image processing techniques to detect humans and recognize their pose
- 4 To Establish communication between the Raspberry Pi and Pixhawk through the MAV Link protocol for Synchronized flight control.
- 5 To Test and validate the system's performance in real flight conditions to evaluate stability, accuracy, and responsiveness.

3.4 Methods and Methodology/Approach to attain each objective

Table 3.4 Methods and Methodology

Objective No.	Statement of the Objective	Method/ Methodology	Resources Utilised
1	literature review for human detection, pose estimation and Image processing Techniques and algorithms	Based on the Available resources and their functionality	- YOLOv8 - OpenCV - ArduPilot Documents
2	drone platform integrated with Pixhawk flight controller and flysky with the Raspberry Pi 5 (16GB).	Interface RPI5 and Pixhawk via USB	Raspberry Pi5, Pixhawk2.4.8, Flysky RC
3	Implement image processing techniques to detect humans and recognize their pose	Real-time pose estimation comparing wrist to shoulder Y-coordinates for distress logic	- YOLOv8-Pose - OpenCV - ArduCam IMX519 16MP
4	communication between the Raspberry Pi and Pixhawk through the MAV Link protocol for Synchronized flight control	Using PyMAVlink to fetch Live latitude, longitude from drone's GPS module	- MAVlink - PyMAVlink - GPS module
5	Test and validate the system's performance in real flight conditions to evaluate stability, accuracy, and responsiveness.	- Telegram BOT API for telegram alerts - VNC for real time video surveillance - Mission planner to observe the drone flight data	- Telegram BOT API - VNC Viewer - Google Maps API - Mission Planner

4. Development

The most critical and crucial key to this chapter is how to maximize the Thrust-to-Weight Ratio to be able to include the bulky computational load (Raspberry Pi 5) without compromising the flight stability or safety.

4.1 Hardware Architecture and Component Selection.

In order to guarantee a well-built platform and a powerful one that could survive outdoors and be able to process its edges, the following particular components as shown in **Table 4.1** have been chosen and integrated:

Table 4.1 Hardware Architecture and Component Selection

Component	Component Specification	Motive
Airframe	F450 Quadcopter Frame	High structural integrity with integrated PCB points.
Propulsion	1000KV BLDC Motors (A2212)	Provides high RPM efficiency for 10-inch propellers.
Speed Control	30A SimonK ESCs	Fast refresh rates (400Hz) optimized for multi-rotor stability.
Power Plant	5200mAh 3S LiPo (35C)	High capacity for extended flight times; 35C burst rating ensures safety.
Flight Brain	Pixhawk 2.4.8	Redundant IMU and barometer for precise GPS
Flight Modes	Flysky i6s	Loiter
Companion Computer	Raspberry Pi 5 (8GB/16GB)	YOLOv8, image processing.

4.2 Engineering Calculations and Propulsion Analysis

A mathematical feasibility study was carried out with ascertaining the possibility of the selection of 1000KV motors used and subsequently 30A ESCs so that the system could be lifted safely before assembly. Budgets of weight are calculated as a portion of segregated weight per unit (cubic meter).^[3]

4.2.1 Weight Budget Estimation (All-Up Weight)

Budgets of weight are determined as a percentage of segregated weight per unit (cubic meter). To begin with, the Total Take-off Weight has been established by adding the mass component of the item:

Table 4.2.1 Weight Budget Estimation (All-Up Weight)

Component Category	Original	New Estimate	Change
Airframe & Landing Gear	300 g	450 g	+150 g
Propulsion (Motors + ESCs)	320 g	440 g	+120 g
Energy (Battery Pack)	400 g	800 g	+400 g
Primary Logic (Pixhawk/GPS)	120 g	120 g	0 g
Companion Logic (Pi 5 + Fan)	100 g	110 g	+10 g
Sundries (Wiring, PDB, Props)	60 g	80 g	+20 g
Total Take-Off Weight	1300 g	2000 g	+700 g

4.2.2 Thrust-to-Weight Ratio Calculation

Agility and stability are critical factors to a drone. A drone that is safe needs a minimum of 2:1 Thrust to weight ratio.

Motor Spec: The 1000KV motor is of a standard A2212 type with 10 inch propellers that provides about 800g of thrust at 100% throttle.^[3]

Total Available Thrust

$$T_{max} = 4 \times 800g = 3200 g$$

Ratio calculation :

$$\text{Ratio} = \frac{T_{max}}{W_{total}} = \frac{3200}{1300} \approx 2.46:1$$

Inference: This is an excellent ratio of 2.46: 1. It will result in the drone flying at about 4045% throttle resulting in plenty of remaining power which can be used during fighting strong winds or fast maneuvers during an emergency.^[3]

4.2.3 Battery C-Rating and Current Safety Check

We have to make sure that the 30A SimonK ESCs and the 35C Battery will not burn in full load. Max Current Draught: Each motor at 100 percent throttle has a draught of about 15A.

$$I_{max_load} = 4 \times 15A = 60 A$$

ESC Capacity:

$$4 \times 30A = 120A \text{ total capacity. Since } 60A < 120A$$

the SimonK ESCs are certainly clear of the danger zone.

4.2.4 Battery Safety

$$I_{battery_max} = \text{Capacity (Ah)} \times \text{C-Rating}$$

$$I_{battery_max} = 5.2 \text{ Ah} \times 35 = 182 A$$

182A can be provided by the battery, however, at best, the motors will pull up 60A. The electric circuit is also safe.

4.2.5 Theoretical Flight Time Calculation

$$\text{Time}_{flight} = \frac{\text{Usable Capacity (Ah)}}{\text{Hover Current (A)}} \times 60 \text{ min}$$

$$\text{Time}_{flight} = \frac{4.16}{17} \times 60 \times 60 \approx 876 s$$

4.3 Hardware Integration and Wiring Strategy

The power circulation was done in a Power Distribution Board (PDB).

- PDB Integration: Battery lead (XT60) is also attached to the PDB. The four SimonK ESCs are peeled off and solder on the pads of the PDB to balance the 11.1 V equally.
- BEC (Battery Elimination Circuit): Because the voltage is 11.1V AC the Pixhawk and Raspberry Pi are unable to read it directly, a 5V/3A BEC is needed to reduce the voltage to the flight controller.^[3]

4.3.1 Motor Mapping

- Motor 1 and 2 Counter-Clockwise (CCW)
- Motor 3 and 4 Clockwise(CW)

Poor wiring in this case can cause the drone to undergo a flip as soon as it takes off.

4.4 AI Vision Pipeline Implementation

The section describes the software logic that will be executed on the Raspberry Pi 5

4.4.1The Detection Logic

The software is in the form of a continuous Loop. The system is programmed so that it helps in filtering out the noise (random movements) and only activates the SOS-like gestures.

System Logic Flow:

Start: Welcome to the Camera and Load YOLOv8 Model.
input Capture Frame (720p resolution).

Process 1 (Detection):

- Load YOLO Inference.
- Condition: Is Class Person and Confidence Above 0.6?
- NO: Return to Start.
- YES: Continue to Pose Estimation.

Process 2 (Skeleton Extraction):

- Get (X, Y) points at RightWrist, LeftWrist, right, and left shoulders.

Process 3 (Geometric Logic):

- Checking: check whether YWrist is smaller than YShoulder (in images, Y is smaller on your way up).
- Condition: Do both wrists go above shoulders?

Process 4 (Temporal Filter):

- Timer Check: Has it taken more than 1.0 seconds to be this true?
- YES: TRIGGER ALERT.

Output: Send the Telegram Alert having GPS location.

4.5 Advantages of Raspberry Pi 5

Past versions with Raspberry Pi 4 were not able to sustain 10 FPS (Frames Per Second). Raspberry Pi 5 offers an important increase in performance, and now it is possible to run the quantized YOLO model with 25-30 FPS, which is vital in real-time tracking when flying.

4.6 Communication System (Telegram Bot)

The communication link does not rely on complex ground station software but uses a standard messaging app for universal accessibility.

API Token: A unique bot token is generated via "BotFather."

- Chat ID: The rescue team's group ID is hardcoded into the Python script.
- Payload: When the "Trigger Alert" function is called, the script executes:

```
Python def send_alert(image, gps_lat, gps_lon);  
caption=f"SOS DETECTED!\nLat:{gps_lat}\nLon:{gps_lon}\nAlt:15m"bot.send_photo(chat_id,  
photo=image, caption=caption);
```

4.7 Validation Scenarios

To ensure the hardware and software work in unison, the following test bench was established:

1. Bench Test (No Propellers):

- Power up system.
- Perform SOS gesture in front of camera. o Pass Criteria: Motors should NOT disarm, but the Telegram notification must arrive within 3 seconds.

2. Vibration Test:

- Run motors at 50% throttle (clamped down).
- Check for "Jello Effect" in camera feed.
- Solution: Soft-mount dampers were added under the Pi Camera to reduce rolling shutter distortion caused by the 1000KV motors.

3. Load Test:

- Hover at 2 meters for 10 minutes.
- Measure battery voltage and motor temperature.
- Pass Criteria: Motors should be warm (not hot), and voltage should be >

5. Results and Discussions

In this project, the AI-Assisted Drone, which should perform the roles of search and rescue teams. With the integration of the CV and the available equipment, the drone searches the site of the victims, detects if they are in distress or they are safe, and sends real-time notifications to the emergency services in remote locations.

The AI-Assisted Drone For Disaster Management was created by considering, selecting and using basic but effective parts, to come up with a prototype that is capable of identifying humans using the aerial perspective, process the information, and relay it to the telegram API BOT.

Since there are no active disaster situations and our drone being just a prototype, the Ramaiah Play Ground was assumed to be a naturally occurred situation, and the drone was deployed to test and validate the situation for the required results.

5.1 MAVlink Connection

The Mavlink Connection is a protocol that is designed for exchanging data between drones, ground station and on board components. In **Figure 5.1** we have established a MAVlink connection between the Raspberry Pi and Pixhawk using a USB and “PyMAVlink” to fetch Live latitude, longitude from drone’s GPS module, we have used Wi-Fi for the communication between the drone and Ground station

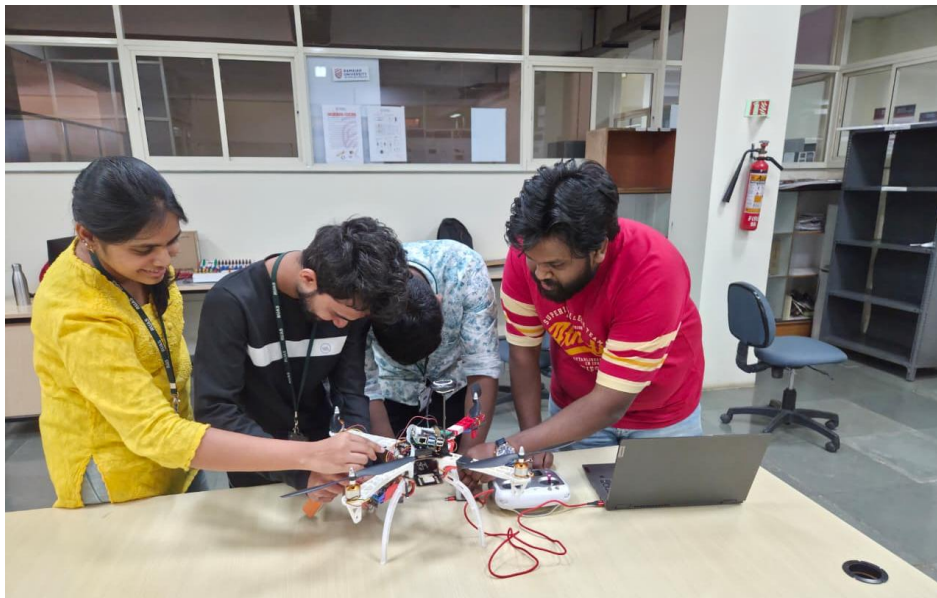
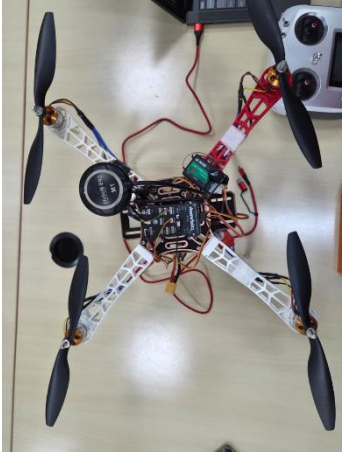
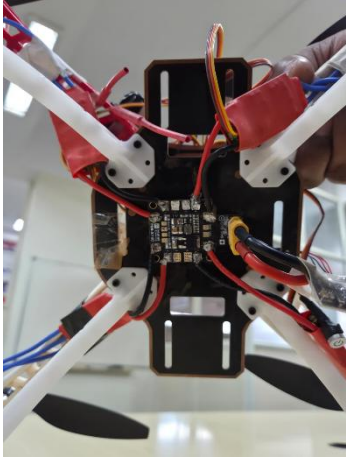


Figure 5.1 MAVlink Connection

5.2 Drone Platform/Prototype

Table 5.2 Drone Platform/Prototype

Figure 5.2.1 Drone Top View • Pixhawk • GPS Module 7m • F450 Frame • Propellers 10 inch • FlySky Rx/Tx	
Figure 5.2.2 Drone Side View • Motors 1000 kV	
Figure 5.2.3 Drone Bottom View • PM Module • PDB Board • ESC	

5.3 Ground Station

The ground station consists of 2 major applications, RealVNC- to have a visual of Real-Time Video Processing, The Mission Planner is used to calibrate, configure, and observe the drone system, here UPD (User Datagram Protocol) is used for fast time streaming of data of the drone as shown in **Figure 5.3**

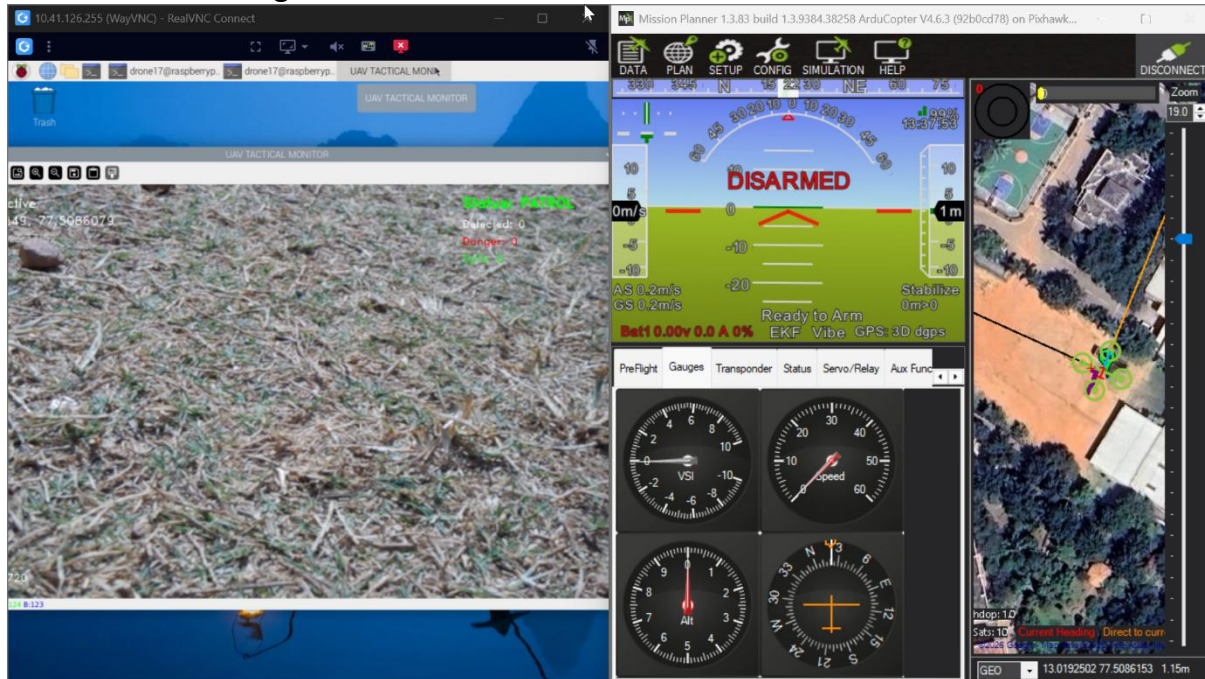


Figure 5.3 Ground Station

5.4 Drone Arming and flight

Drone arming is a basic safety check before the flight, it performs checks on compass, GPS, gyroscope before allowing the motors to arm as shown in **Figure 5.4 (A)** ensuring safety starting the motors with a low rpm throttle as shown in **Figure 5.4 (B)**



Figure 5.4 (A) Drone Arming



Figure 5.4 (B) Arming check low RPM Take off

5.5 Human Detection

The Raspberry Pi is connected with a camera module “ArduCam” and the human detection is based on the YOLOv8 pose estimation model in **Figure 5.5.1**, in **Figure 5.5.2** we can see the detection being observed in the realVNC, and the basic detection is performed with an ordinary camera module from the second floor(approximate 9m), RUAS building.



Figure 5.5.1 RPI&CAM

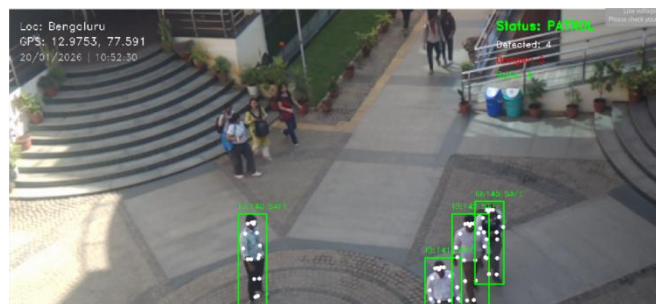


Figure 5.5.2 Human Detection

5.6 Drone Flying/testing

The drone is deployed in an empty ground, assuming that the area is a natural disaster, here in **figure 5.6**, the drone is flown to an approximate height of 12 meters



Figure 5.6 Drone Flying/testing

5.7 Real-Time Human Detection From the drone

Assuming the Ramaiah Play Ground is a Natural Disaster situation, a group of 8 people are present to test and validate the system as shown in **Figure 5.7 (A)** The raspberry pi is mounted on the drone, and is connected to the Pixhawk via MAVlink, the data is processed and is shared between the 2 devices, producing GPS coordinates, time and date **Figure 5.7 (B)** we can see that the human detection is taking place in our assumed natural disaster with proper coordinates and visuals with approximate 6m height



Figure 5.7 (A) people are present to test and validate the system

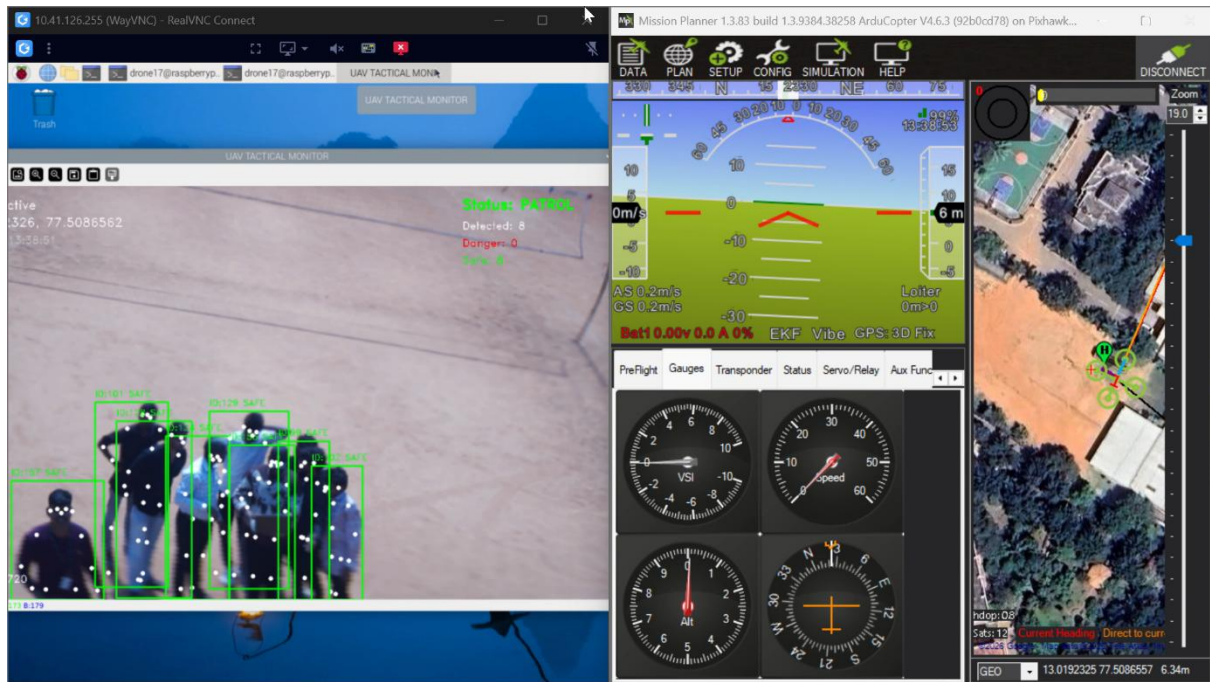


Figure 5.7 (B) Number of Humans Detected and are Safe



Figure 5.7(C) Humans Detected and in Danger

As shown in **Figure 5.7(B)**, the status of the drone is on patrol as all the detected people in the frame are safe, where as in **Figure 5.7(C)** there is one detected as danger amongst 7 detected people. Therefore 1 out of 8 Human have not been detected making our system 90% accurate.

5.8 Telegram Alert/Output

Table 5.8 Telegram Alert/Output

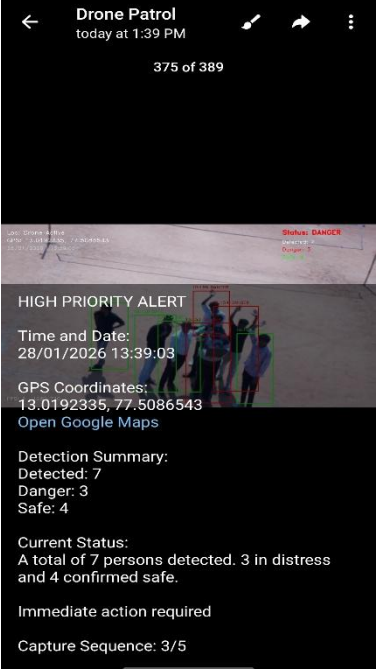
Figure 5.8.1 Telegram Alert This figure shows the alert sent to the user/ ground station. With a message including • Time and date • GPS coordinates • Google Map Link • Detection Summary • Status • Capture sequence (first 5 frames)	 The screenshot shows a Telegram chat interface with a message titled "Drone Patrol" sent "today at 1:39 PM". The message content includes: - A video frame showing a group of people with bounding boxes. A red box highlights one person with the text "Status: DANGER". HIGH PRIORITY ALERT Time and Date: 28/01/2026 13:39:03 GPS Coordinates: 13.0192335, 77.5086543 Open Google Maps Detection Summary: - Detected: 7 - Danger: 3 - Safe: 4 Current Status: A total of 7 persons detected. 3 in distress and 4 confirmed safe. Immediate action required Capture Sequence: 3/5
---	--

Figure 5.8.2 Google Link

This figure shows the Google link sent with the GPS Coordinates

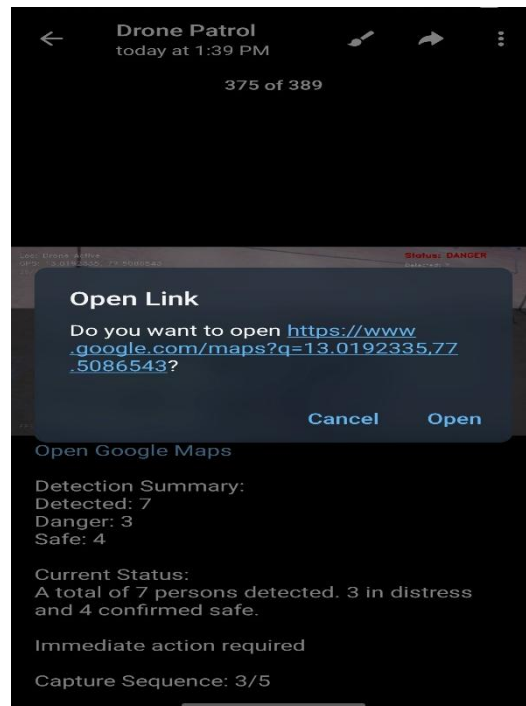
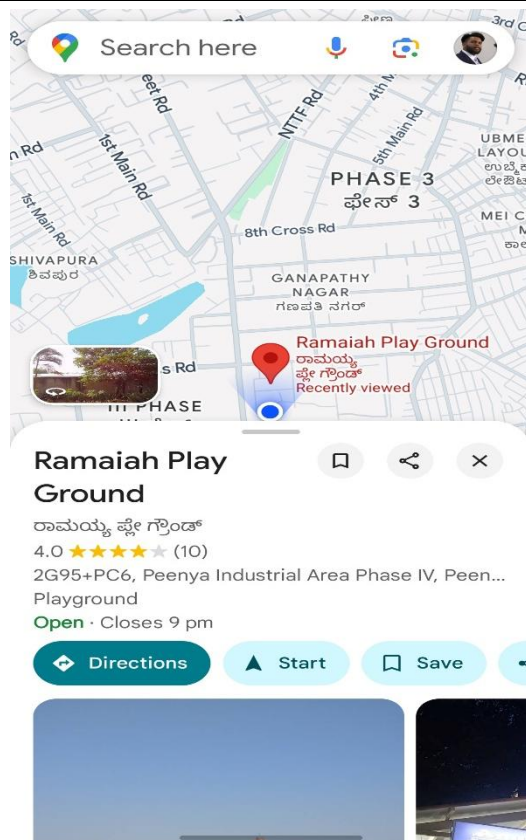


Figure 5.8.3 Google Maps

In this figure, PyMAVlink Fetches the raw coordinates from the Pixhawk,
The same coordinates are formatted with a string "[//www.google.com/maps?q=coordinates](https://www.google.com/maps?q=coordinates)." This string is later attached with the Telegram alert for the user/ rescue team to open the maps and get the location of the victim



6. Project Costing

Building a custom drone requires a meticulous balance of high-performance electronics and structural reliability. The Table 6.1 shows the hardware list outlines the essential components needed for this project

Table 6.1 Project Costing

SL.No	Components	Quantity	Costing (rs)	Source
1.	Raspberry pi	1	16500	Robu.in
2.	Pixhawk	1	7000	Robocraze.com
3.	Flysky fsi6 tx and rx	1	6000	Robu.in
4.	PM module	1	500	Robu.in
5.	F450 frame	1	800	Robu.in
6.	PDB board	1	270	Robocraze.com
7.	ESCs	4	1200	Robu.in
8.	Motors 1000kv	4	1480	Robocraze.in
9.	Battery 5200mah 11.1v , 35C	1	4000	Local
10.	Camera (ardu cam)	1	2500	Robocraze.com
11.	GPS module 7M	1	1600	Robu.in
12.	Propellers	10	1000	Robu.in
13.	Other accessories	-	2000	mixed
14.	TOTAL	-	44,850	-

The selection of hardware for this drone project focuses on creating a platform by integrating high-level computing with robust flight control. At the core of the system is the Raspberry Pi, which serves as a companion computer to handle image processing. This works in alongside with the Pixhawk flight controller, the primary hardware responsible for stabilizing the aircraft and executing flight commands based on data from its internal sensors and the 7M GPS module. To facilitate manual control and communication, the Flysky FS-i6 transmitter and receiver set provides a reliable link between the pilot and the drone's processing units.

The structural and power systems are designed to support these electronics while maintaining flight efficiency. The F450 frame provides the physical skeleton, while a Power Distribution Board (PDB) and Power Module (PM) manage the energy flow from the 5200mAh 11.1V battery. The PDB ensures that high current is distributed evenly to the four Electronic Speed Controllers (ESCs), which in turn regulate the speed of the 1000kv motors. These motors drive the propellers to generate the necessary thrust for lift. Finally, the build is completed with various accessories, such as wiring and connectors.

7. Conclusions and Directions

7.1 Conclusion

AI-ASSISTED DRONE FOR HUMAN DISTRESS DETECTION which is a prototype, that achieved a 90% classification accuracy in from 6 to 12 meters altitude, and 76% at 16 meters during testing. Unlike traditional drone-based detection which relies on observing data manually, this system utilizes YOLOv8-Pose estimation to analyse 17 human key points. This allows the drone to ignore camera motion and focus solely on the victim's gestures. While Cloud-based systems may offer slightly higher accuracy, our onboard implementation on the Raspberry Pi 5 ensures a critical advantage in speed, providing immediate GPS coordinates via Telegram.

7.2 Comparison Between existing model vs Our Prototype

Table 7.2 comparison between S. L. Krishna's Existing system^[5] and our prototype

Feature	S. L. Krishna et al. (2019)	Our Prototype (2026)	Advancement Type
Detection Engine	Standard CNN / RGB Detection	YOLOv8-Pose Estimation	Precision Upgrade: Moves from simple "object box" to "skeleton analysis."
Accuracy	"Moderate" (Altitude Dependent)	90% Consistent in 12meters, 80% in 16meters	Reliability: Improved key point tracking reduces distance-based error.
Intelligence	Binary Human Detection	Safe vs. Distress Segregation	Functional Logic: Adds behavioural context to the detection.
Hardware	2019-era Flight Comp.	Raspberry Pi 5, Pixhawk	Processing Power: Real-time with AI integration and flight control
Alerting System	LED Screen / Local Display	Telegram Bot + Google Maps Link	Connectivity: Remote, instant dispatch to rescuers' smartphones.
Visual Monitoring	Local Video Storage	Real-Time VNC Tactical HUD	Situational Awareness: Live "Pilot-in-the-loop" visual verification.

7.3 Accuracy

The system maintains 90% accuracy up to 12 meters and drops to 76% between 12 and 16 meters. At this range, Key point Jitter occurs. The AI might see a person but struggle to pin the exact location of the wrists, leading to more missed signals.

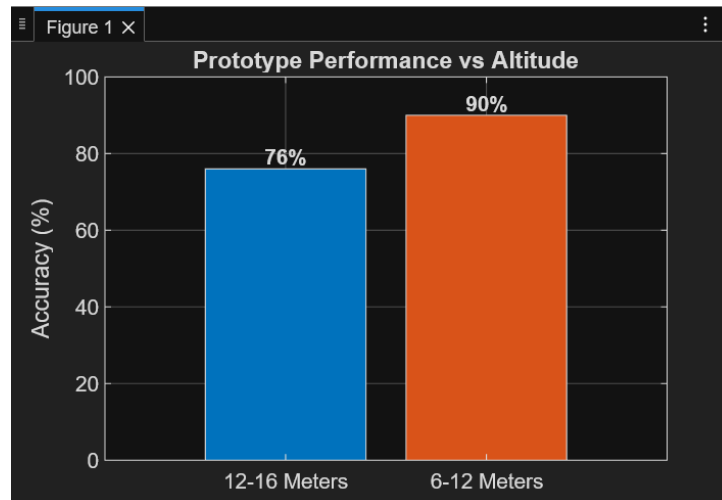


Figure 7.4 Prototype Performance vs Altitude

7.4 Suggestions for future directions

While This Project successfully developed a functional Drone for Human Distress Detection There are several areas that can be explored to make improvements

- (LoRa) Offline Communication: The current reliance on the Telegram API requires a stable Wi-Fi network, which is often unavailable in remote disaster zones. Future versions should incorporate LoRa (Long Range) radio modules to transmit GPS coordinates and distress status directly to a handheld ground station.
- Multi-Spectral Imaging and Thermal Fusion: Currently, the system relies on standard RGB sensors, which are ineffective in low-light, night-time, or heavy smoke conditions. Integrating a thermal camera alongside the existing vision system would allow for "Heat Signature Detection." This would enable the drone to locate victims through dense foliage or at night, significantly increasing the success rate
- Multi-Gesture Recognition: While the "Hands-Up" logic is a highly effective distress signal, expanding the AI's "vocabulary" to include SOS signalling, pointing or other distress related gestures would provide more context to rescuers.

References

1. Caputo, S., Castellano, G., Greco, F., Mencar, C., Petti, N., Vessio, G. (2022). Human Detection in Drone Images Using YOLO for Search-and-Rescue Operations. In: Bandini, S., Gasparini, F., Mascardi, V., Palmonari, M., Vizzari, G. (eds) AlxIA 2021 – Advances in Artificial Intelligence. AlxIA 2021. Lecture Notes in Computer Science(), vol 13196. Springer, Cham. https://doi.org/10.1007/978-3-031-08421-8_22
2. Dong, C. and Du, G., 2024. An enhanced real-time human pose estimation method based on modified YOLOv8 framework.
3. Aliane, Nouridine. 2024. "A Survey of Open-Source UAV Autopilots" Electronics 13, no. 23: 4785. <https://doi.org/10.3390/electronics13234785>
4. You Only Look Once: Unified, Real-Time Object Detection Joseph Redmon, Santosh Divvala, Ross Girshick, Ali Farhad University of Washington , Allen Institute for AI, Facebook AI Research
5. S. L. Krishna, G. S. R. Chaitanya, A. S. H. Reddy, A. M. Naidu, S. S. Poorna and K. Anuraj, "Autonomous Human Detection System Mounted on a Drone," 2019 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET), Chennai, India, 2019, pp. 335-338, doi: 10.1109/WiSPNET45539.2019.9032876. keywords: {Cameras;Telemetry;Sensors;Drones;Conferences;Trajectory;Unmanned Aerial Vehicle (UAV);Natural Disaster;Drone;Flight Controller;Human Detection},

Appendix

Flow Chart of Human Distress Detection:

